

CURRICULUM VITAE

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Professional Address:
Cystic Kidney Disorders Unit
Division of Genetics and Cell Biology
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EDUCATION

- PhD** University Vita-Salute San Raffaele, Molecular Medicine May 2022
Dissertation: “Identification of genetic variants contributing to pain-related symptoms in familial and sporadic patients affected by peripheral neuropathy”
Curriculum: Experimental Neurology
- MS** Bharati Vidyapeeth Deemed University, Bioinformatics May 2016
Thesis: “Comparison of RNASeq analysis protocols using *Cochliobolus Sativus* transcriptome data”
Advisor: Dr. Narendra Kadoo
- MS** University of Exeter, Biotechnology & Enterprise September 2012
Thesis: “SynBIn: A synthetic biochemical “insulator” as a new BioBrick device for synthetic biology”
Advisor: Dr. Nicholas Harmer
- BS** Oregon State University, Bio-Engineering June 2010
Graduated Cum Laude
Thesis: “Mass Transfer Effects on Cell Growth and Protein Production”
Advisor: Dr. Joseph McGuire

HONORS AND AWARDS

- Marie Skłodowska-Curie Early Stage Researcher** 2018
PhD project was funded under Marie Skłodowska-Curie Actions
- Provost International Scholarship Award** 2006
Scholarship is given to the incoming students at Oregon State University based on their merit.

EXPERIENCE

Cystic Kidney Disorders Unit, San Raffaele Hospital, Milan, Italy 2025 to present

Project Manager – Scientific Projects

- Managed scientific initiatives including 1 European-funded project, Cilia2027 international conference, and an Italian national consortium
- Coordinated multidisciplinary teams across institutions and international partners
- Planned timelines, milestones, and deliverables in line with funding and institutional requirements
- Served as main liaison between researchers, and administrative stakeholders
- Oversaw reporting, documentation, and progress tracking

Human Genetics of Neurological Disorders Unit, San Raffaele Hospital, Milan, Italy
2022 to 2025

Junior Post Doctorate Fellow

- Automated next-generation sequencing pipelines by using Snakemake including Whole exome, Whole genome, RNASeq, miRNA-Seq, Single-Cell Seq analysis
- Data management and handling
- Independent and paired development on bugs and features of pipeline work flow system
- Creating and validating computation pipelines
- Providing bioinformatics support
- Devised source version control by using git and bitbucket
- Created a clinical database for laboratory usage
- Handled various variant interpretation and classification projects (Processed more 200 samples)

Early Stage Researcher, University Vita-Salute San Raffaele, Milan, Italy 2018-2022
Supervisor: Dr. Federica Esposito

PhD Title: “Identification of genetic variants contributing to pain-related symptoms in familial and sporadic patients affected by peripheral neuropathy”

- Written Perl /python/R/shell scripts to automate next-generation sequencing pipeline
- Independent and paired development on bugs and features of pipeline work flow system
- Devised source version control by using git and bitbucket
- Created web survey page by using LimeSurvey tool
- Created a structural variation pipeline using Snakemake
- Variant interpretation and classification according to ACMG Guidelines on 150 neurological samples

MedGenome Private Limited, Bengaluru, India 2016 to 2018

Bioinformatics Analyst

- Written Perl /R/python/shell scripts to automate next-generation sequencing pipeline
- Devised source version control and Jenkins in several projects

- Created dashboards in R Shiny for clinical data visualization
- Expertise in regression testing, smoke testing, user acceptance testing, performance testing, and functional testing and backend testing
- Provided timely and accurate responses to data requests, consulting with client for accuracy of requirements of request, and ensuring deadline request was met
- Experience in creation of test cases, test plans and test release documents
- Leading team of 3 people, responsible for issues tracking, work assignment.

IVF Craft Clinic, Mumbai, India

2012 to 2014

Business Development Manager

- Building client relationships and delivering presentations or proposals,
- Manage a team to explore and actively pursue growth opportunities overseas
- Establish, maintain, and strengthen appropriate internal control safeguards
- Assisting an embryologist in performing assisted reproductive technologies (ART) to help with infertility
- Maintaining an understanding of human reproductive biology, embryology, infertility and ART;
- Having knowledge of and complying with quality control, ethical issues, and regulations surrounding gamete and embryo handling; care and maintenance of equipment

Oregon State University, Corvallis USA

Summer 2009

Research Intern

- Researched effects of different substrates on Microbial Fuel Cell; learned to operate microbial electrolysis cell
- Designed and constructed microbial fuel cells (MFC) using polyethylene cubes, carbon anodes, and platinum coated cathodes.
- Assisted in isolating bacterial colonies and testing electricity generation capabilities.
- Performed DNA extraction, and PCR gel & electrophoresis techniques to analyze bacterial community.
- Enriched bacteria with different anaerobic and aerobic conditions, and measured hydrogen gas production.
- Worked with graduate students to improve the design and efficiency of Microbial Fuel Cell.
- Gained familiarity with various lab measurement instrumentation and devices, such as making various mediums

PROFESSIONAL TRAINING

DI.V.A.L Toscana Srl, PhD Intern, December 2019

- Learned how to use cell culture and preparation for patch-clamp
- Learned how to operate Automated patch clamp electrophysiology
- Learned how to do patch-clamp recording of a single cell

University of Maastricht, PhD Intern, Neurology Department, October 2019

- Created a R program for processing multiple annotation files
- Learned how to prioritize genetic variants in different types of samples

Fondazione IRCCS Istituto Neurologico Carlo Besta, PhD Intern, Neuroalgology Unit, July 2019

- Created a Whole exome sequencing pipeline analysis and automated using Snakemake
- Performed Polygenic Risk Score calculation using PRSice and PLINK tool

PUBLICATIONS

Journal Publications

Kaalindi Misra, Aishwary Nerkar, Ferdinando Clarelli, Melissa Sorosina, and Federica Esposito. "Long Non-Coding RNAs in Multiple Sclerosis—Differential Expression and Functional Implications." *Genes* 16, no. 11 (2025): 1327.

Antonino Giordano, Arianna Giliberti, Ferdinando Clarelli, **Kaalindi Misra**, Elisabetta Mascia, Melissa Sorosina, Giulia Visentin et al. "Sex hormone-related factors and the risk of PIRA in women with multiple sclerosis." *Journal of Neurology, Neurosurgery & Psychiatry* 96, no. 7 (2025): 686-689.

Antonino Giordano, Ferdinando Clarelli, Béatrice Pignolet, Elisabetta Mascia, Melissa Sorosina, **Kaalindi Misra**, Laura Ferrè et al. "Vitamin D affects the risk of disease activity in multiple sclerosis." *Journal of Neurology, Neurosurgery & Psychiatry* 96, no. 2 (2025): 170-176.

Ferdinando Clarelli, Andrea Corona, Kimmo Paakkonen, Melissa Sorosina, Alen Zollo, Fredrik Piehl, Tomas Olsson, **Kaalindi Misra**, et al. "Pharmacogenomics of clinical response to Natalizumab in multiple sclerosis: a genome-wide multi-centric association study (vol 271, pg 7250, 2024)." *Journal of Neurology* 272, no. 6 (2025).

Kaalindi Misra, Milena Ślęczkowska, Silvia Santoro, Monique M Gerrits, Margherita Marchi, Erika Salvi, Hubert J M Smeets, Janneke G J Hoeijmakers, Martinelli Boneschi Filippo Giovanni, Massimo Filippi, Giuseppe Lauria Pinter, Catharina G Faber, and Federica Esposito, "Broadening the genetic spectrum of painful small fibre neuropathy through whole exome study in Early-Onset cases", *International Journal of Molecular Sciences* 25, no. 13 (2024): 7248.

Elisabetta Mascia, Valentina Nale, Laura Ferrè, Melissa Sorosina., Ferdinando Clarelli, Alice Chiodi, Silvia Santoro, Antonino Giordano, **Kaalindi Misra**, Miryam Cannizzaro, Lucia Moiola, Vittorio Martinelli, Luciano Milanesi, Massimo Filippi, Ettore Mosca and Federica Esposito, "Genetic contribution to medium-term disease activity in multiple sclerosis," *Molecular Neurobiology* (2024): 1-13.

Antonino Giordano, Béatrice Pignolet, Elisabetta Mascia, Ferdinando Clarelli, Melissa Sorosina, **Kaalindi Misra**, Florence Bucciarelli et al. "DNA Methylation in

the Anti-Mullerian Hormone Gene and the Risk of Disease Activity in Multiple Sclerosis." *Annals of Neurology* (2024).

Valentina Guarino, Jessica Gliozzo, Ferdinando Clarelli, Béatrice Pignolet, **Kaalindi Misra**, Elisabetta Mascia, Giordano Antonino et al. "Intrinsic-Dimension Analysis for Guiding Dimensionality Reduction in Multi-Omics Data." *Bioinformatics*, p. 243-251. (2023).

Milena Ślęczkowska[§], **Kaalindi Misra**[§], Silvia Santoro, Monique M. Gerrits, and Janneke GJ Hoeijmakers. "Ion Channel Genes in Painful Neuropathies." *Biomedicines* 11, no. 10 (2023): 2680. [§]First authors

Journal Papers in Review

Antonino Giordano, Pernilla Stridh, Paolo Preziosa, Marco Pisa, Melissa Sorosina, Elisabetta Mascia, Silvia Santoro, **Kaalindi Misra**, Ferdinando Clarelli, Laura Ferré, Maria Needhamsen, Ali Manouchehrinia, Miryam Cannizzaro, Thomas Moridi, Klementy Shchetynsky, Russell Ouellette, Adil Harroud, Elisabeth Sandberg, Subita Balaram Kuttikkatte, Fredrik Piehl, Lars Alfredsson, Jan Hillert, Tomas Olsson, Lars Fugger, Kate Attfield, Tobias Granberg, Maja Jagodic, Gabriele DeLuca, Mara Rocca, Massimo Filippi, Ingrid Kockum, Federica Esposito, "Genetic variation in HIF1A is associated with smoldering inflammation and disease progression in Multiple Sclerosis." *medRxiv* (2024): 2024-03.

Conference Papers

Melissa Sorosina, **Kaalindi Misra**, Ferdinando Clarelli, Antonio De Simone, Antonino Giordano, Elisabetta Mascia, Maria Assunta Rocca, Massimo Filippi, and Federica Esposito. "Age-dependent accumulation of somatic mitochondrial DNA mutations is different between multiple sclerosis progression phenotypes." *Multiple Sclerosis Journal*, 2025, pp. 13-13.

Antonino Giordano, **Kaalindi Misra**, Michele Surano, Aishwary Nerkar, Ferdinando Clarelli, Melissa Sorosina, Elisabetta Mascia, Martina Absinta, Massimo Filippi, and Federica Esposito. "Somatic mutations profiling of immune cells in Multiple Sclerosis lesions." *Multiple Sclerosis Journal*, 2025, pp. 43-43.

Matteo Missaglia, Melissa Sorosina, Elisabetta Mascia, **Kaalindi Misra**, Antonino Giordano, Ferdinando Clarelli, Maria Assunta Rocca, Federica Esposito, and Massimo Filippi. "Differentially expressed miRNAs in Multiple Sclerosis patients stratified for NEDA status." *Multiple Sclerosis Journal*, 2025, pp. 953-954.

Chiara Zanetta, Antonino Giordano, Francesco Laserra, Federica Esposito, Melissa Sorosina, Elisabetta Mascia, **Kaalindi Misra**, Vittorio Viti, Maria Assunta Rocca, and Massimo Filippi. "Environmental risk factors in subjects with radiologically

isolated syndrome: a case-control study." Multiple Sclerosis Journal, 2025, pp. 569-569.

Melissa Sorosina, **Kaalindi Misra**, Ferdinando Clarelli, Elisabetta Mascia, Antonino Giordano, Sara Scire, Laura Ferré, Miryam Cannizzaro, Michela Crotti, Daniele De Ritis, Francesca Maltecca, Massimo Filippi, Federica Esposito, "Genetic determinants of mitochondrial impairment in primary progressive multiple sclerosis", Multiple Sclerosis Journal, Oct. 2023, p. 804

Antonino Giordano, **Kaalindi Misra**, Ferdinando Clarelli, Melissa Sorosina, Elisabetta Mascia, Silvia Santoro, Laura Ferré, Miryam Cannizzaro, Lucia Moiola, Vittorio Martinelli, Massimo Filippi, Federica Esposito, "Use of oral contraceptive, menopause and pregnancy before Multiple Sclerosis diagnosis are associated with more favorable disease course", Oct. 2023, p. 179-180

Antonino Giordano, Melissa Sorosina, Elisabetta Mascia, Beatrice Pignolet, Silvia Santoro, Ferdinando Clarelli, Laura Ferré, **Kaalindi Misra**, Miryam Cannizzaro, Roland Liblau, Massimo Filippi, Federica Esposito, "Vitamin D is causally associated with the No Evidence of Disease Activity-3 status in Multiple Sclerosis", Oct. 2023, p. 295-296

Valentina Guarino, Jessica Gliozzo, Ferdinando Clarelli, Béatrice Pignolet, **Kaalindi Misra**, Elisabetta Mascia, Giordano Antonino, Silvia Santoro, Laura Ferré, Miryam Cannizzaro, Melissa Sorosina, Roland Liblau, Massimo Filippi, Ettore Mosca, Federica Esposito, Giorgio Valentini, Elena Casiraghi, "Intrinsic-Dimension analysis for guiding dimensionality reduction and data-fusion in multi-omics data processing," Proceedings of the 16th International Joint Conference on Biomedical Engineering Systems and Technologies, Feb. 2023, p.243-251.

Valentina Nale, Ferdinando Clarelli, Noemi Di Nanni, Laura Ferré, Elisabetta Mascia, Silvia Santoro, Matteo Gnocchi, Marco Moscatelli, Martina Tosi, **Kaalindi Misra**, Melissa Sorosina, Luciano Milanese, Massimo Filippi, Federica Esposito, Ettore Mosca, "Interactome-based multi-omics study of molecular networks implicated in disease activity in Multiple Sclerosis," European Journal of Human Genetics, Apr. 2022, p. 317-318.

Melissa Sorosina, Silvia Santoro, Béatrice Pignolet, **Kaalindi Misra**, Antonino Giordano, Elisabetta Mascia, Ferdinando Clarelli, Miryam Cannizzaro, Laura Ferré, Ettore Mosca, Roland Liblau, Massimo Filippi, Federica Esposito, "Analysis of miRNA profiles in active and non-active multiple sclerosis patients", European Journal of Neurology, Jul. 2022, p. 644

Kaalindi Misra, Silvia Santoro, Andrea Zauli, Margherita Marchi, Erika Salvi, Raffaella Lombardi, Daniele Cazzato, Filippo Martinelli Boneschi, Massimo Filippi, Federica Esposito, Giuseppe Lauria; "Genetic study of Italian families affected by

small fibre neuropathy identified variants in predisposing pain phenotype,” Journal of Neurological Sciences, Oct. 2021, p. 225.

Vivek Gopalan and **Kaalindi Misra**, “An automated sample allocation system for multiplexed Next-Generation Sequencing,” Canadian Journal of Biotechnology, Dec. 2017, p. 241.

PRESENTATIONS AND INVITED LECTURES

Workshop, “Applications of Structural Chemistry,” Bioinformatics, 11-02-2016.

Workshop, “Introduction to Bioinformatics,” Bioinformatics, 07-01-2016.

PROFESSIONAL AFFILIATIONS

International Association for the Study of Pain, 2021-2025
Communication committee member.

LANGUAGES

English: Native Language

Italian: Intermediate Listener, Novice Speaker

COMPUTER SKILLS

Programming: C, C++ Oracle, HTML, Perl, R, Python and JAVA Script

Applications: Snakemake, make, GitHub, Jenkins, Microsoft Office Suite